

ECO 364—THINKING STRATEGICALLY

Online Course

Summer Session I: May 28–July 6, 2019

Department of Economics
Stony Brook University

1 Statement about Course Delivery/Modality

This is an asynchronous, online course, delivered in the BlackBoard Learning Management System (LMS). Students must be mindful of all course expectations, deliverables and due dates. All assignments and course interactions will utilize internet technologies. See the section on *Technical Requirements* for more information.

The official time zone for this course is the **Eastern Time Zone (EST)**¹. All deadlines will be indicated using EST time, so please be mindful of that in case you are in another time zone.

2 Instructor & Contact Information

Instructor: Alejandro Melo Ponce

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Office: N604, Social and Behavioural Sciences Building.

Office Hours: By appointment in person or via phone/video conference.

My preferred method of contact is via email. I will strive to respond emails as soon as possible but allow for 24 hours for a response. Please use your Stony Brook University email when contacting me as that is the preferred method of contact for the institution. When sending me an email, *please include in the subject line the keywords ECO364*, since that will insure that I will be able to locate your email and respond to it quickly.

3 Course Prerequisites

C or higher in ECO 303 (Intermediate Microeconomic Theory).

4 Course Description

The manager of the Hermon Ski resort must choose the operation strategy of the site for the coming year: when to start the advertising campaign, when to open the site, and so on. His decision depends on the weather forecast and other factors. Although the manager does not know when the snow accumulation will enable skiing, the snow does not try to outsmart the manager nor does it compete with him. Decision making under uncertainty by a single decision maker is the subject matter of decision theory. Broadly speaking, is the study of the reasoning underlying a decision maker's choices.

Compare the decision problem this manager faces to Google's problem when it needs to choose their strategy for its web search engine. In this problem, Microsoft, with its search engine Bing, replaces the snow. But unlike the snow, Microsoft does compete with Google. Therefore, when choosing Google's strategy, Google needs to consider and analyze the strategy adopted by Microsoft. But when the latter chooses its strategy, it

¹US East Coast time.

surely takes into account Google's analysis of the problem. Therefore, Google has to think how Microsoft is analyzing its decision even before it itself makes it.

This is the sort of problem one faces when analyzing decision-making in situations where several decision makers interact. Decision-making in such circumstances has unique characteristics which are the result of the influence that each individual's decision exerts on all other decision makers. Decision-making in interactive situations is the subject matter of game theory. The types of interactions to which this theory applies are many and varied. Individuals engaged in electing a chair person of an organization, employers and employees negotiating terms of employment, competing firms, parties in the parliament, or countries in conflict are all examples on which game theory can shed light.

4.1 Course Objectives

- Be able to outline examples of strategic interactions.
- Introduce the students to some of the tools and frameworks of Game Theory and use them to analyze some real world issues.
- Introduce the students to some applications of Game Theory: Industry competition, committee decisions, voting, network traffic analysis, auctions and markets.

4.2 Main Topics

1. Decision under uncertainty, or how to make choices under risk.
 - (a) Two Puzzles on Decision making under uncertainty.
 - (b) Modeling uncertainty.
 - (c) Expected utility, the starting point.
2. Dominant strategies, or when it is possible to disregard competitors & Strategic Dominance relationship
 - (a) The prisoner's dilemma—why does he confess?
 - (b) The tragedy of the commons and other disasters.
3. Strategic equilibrium
 - (a) Is the more the better?
 - (b) Strategic voting in the board of directors.
 - (c) How to think about what my competitor thinks about what I think about him...
 - (d) The value of information under strategic conflict; Is it possible to disregard information in interactive situations?
 - (e) What will happen to the traffic if we close Sixth Avenue? One has to see to believe it.
 - (f) Traffic equilibrium and its application.
 - (g) Mixed Strategies.
 - (h) Where should I place my gas station—location games.
4. Dynamic competition
 - (a) Identifying strategic alternatives.
 - (b) The backward induction principle.
 - (c) Perfect equilibrium, or your threats do not frighten me.
5. Auctions.
 - (a) First and second price auctions

- (b) English and Dutch Auctions.
 - (c) Third price auction
 - (d) Common and private values.
 - (e) The winner's curse.
 - (f) Sequential auctions.
 - (g) Merger and acquisition.
 - (h) Tender offers.
6. Stable matches in two-sided markets
- (a) How the hospital-residents market in the US collapsed.
 - (b) Are men better off wooing women or are they just making fools of themselves?
 - (c) The Gale-Shapley matching algorithm.

5 Course Structure

The course is divided into six modules, one for every week of the Summer Session I. During the first module we will cover topic 1, decision under uncertainty. The second module corresponds to topic 2, Dominant Strategies and the Strategic Dominance relationship, and roughly half of topic 3, Strategic Equilibrium. In the third module we will finish covering topic 3, Strategic Equilibrium. The fourth module corresponds to the topic of Dynamic Competition and Auctions. In the fifth module we will finalize the topic of auctions and will start with stable matches in two-sided markets. Finally, in the sixth module we will finalize two-sided markets.

In each module, you will have to complete two different class of assignments: participate in discussion forums and a quiz for the respective module. In addition to this, after all modules are completed, you will have a final exam that will cover all the material from the course. Please see below for more information.

6 Bibliography & Course Material

There are no required textbooks. The lecture notes, quizzes, discussion forums and other content provided on Blackboard are in principle sufficient for the course. All of these material will be posted on BlackBoard <https://blackboard.stonybrook.edu/>.

However, for some topics I will recommend some additional reading from the following source in case you want to reaffirm some concepts:

- Watson, J. *Strategy: An Introduction to Game Theory*, W. W. Norton & Company, (2013).

Finally, the following books are highly recommended for “bedtime reading” while you are taking the course:

- Brandenburger, B. and Nalebuff, B. *Co-Opetition: A Revolution Mindset The Combines Competition and Co-operation*, (1997).
- Dixit, A. and Nalebuff, B. *Thinking Strategically*, Norton (1991).
- Dixit, A. and Nalebuff, B. *The Art of Strategy*, (2008).

You can find the previous sources on the Amazon Stony Brook Bookstore, but again they are not required.

7 Course Assessment and Grading Scale

In this course, you will be assessed according to the following grading policy:

ASSIGNMENTS	% OF GRADING
Attendance, Participation and Discussion	35%
Module Quizzes (end of each module, see details below)	30%
Online Final Exam (July 6th–July 7th, see details below)	35%

7.1 Assignment Description

7.1.1 Attendance, Participation and Discussion (35%)

Attendance Attendance in an online course means **logging into the Blackboard on a regular basis and participating in the all of activities that are posted in the course**. This online course requires that you log in throughout the week for the duration of the course; each weekly module will be available as we progress throughout the term and you will be responsible for logging and understanding the module's deliverables.

Participation Participation in this course is comprised of discussion forums that are required and graded. There are a total of 7 **graded discussion forums**, one for each module and the mandatory Introduction forum in the welcome week, each one of them worth 5 points.

The first week is the only one in which you will have two forums, an Introduction forum and that week's forum for the module topic. In the introduction forum you will need to introduce yourself to the rest of the class. The introduction forum is worth 5 points.

For each module, you will be required to post an original and thoughtful response to the topic of that module's forum. For some modules you will have to post an initial response before you start studying the material for that week's module and then post another comment after you have studied the material. In all discussion forums you must also post at least two responses to two different posts by your colleagues. I will be more specific in your tasks for each module's discussion forum in its BlackBoard description.

I will grade the weekly module forums according to the following rubric:

Score	Knowledge	Understanding	Writing Skills	Application/Analysis	Peer responsiveness
5	Comprehensive, in-depth and wide ranging	Outstanding ability to grasp concepts and Relate theory to practice	Excellent mechanics, sentence structure, and organization	Grasps inner relationship of concepts, Excellent use of a wide range of supporting material	Demonstrates an exceptional ability to analyze and synthesize student work, asks meaningful extending questions
4	Up to date and relevant.	High level of ability to conceptualize essential ideas and relate theory to practice.	Significant mechanics, structure and organization.	Demonstrates the ability to analyze and synthesize, independent analysis, good use of a range of supportive material.	Demonstrates good ability to analyze other student work, and ask meaningful extending questions.
3	Relevant but not comprehensive.	Some ability to conceptualize essential ideas and relate theory to practice.	Some grammatical lapses, uses emotional response in lieu of relevant points.	Informed commentary with some evidence of genuine analysis; some supportive materials used.	Some ability to meaningfully comment on other student work and ask bridging questions.
2	Limited superficial knowledge/response.	Limited ability to draw out concepts and relate theory to practice.	Poor grammar, weak communication.	Some observations, some supportive evidence used.	Lack of ability to comment on other students work and ask meaningful questions.
1	Little relevance/some accuracy.	Minimal awareness that external concepts exist.	Lack of clarity, poor presentation of thinking.	Lacks evidence of critical analysis, poor use of supportive evidence.	Demonstrates poor ability to comment on other student work, asks no meaningful questions.
0	Did not participate in forum.				

Participation in the weekly forums is of extreme importance in this course, so no make-ups are available for the discussion forums.

Discussion You are also responsible to **comment a few of your colleagues' posts**. You are to conduct yourself professionally every time you express yourself-whether it's through the written or spoken word.

7.1.2 Module Quizzes (30%)

Each module ends with a quiz which evaluates your mastery of that module's material. There will be 6 quizzes in total, each one worth 5 points towards your final grade, for a total of 30 points. The quizzes open on the Friday corresponding to that week's module at 12:00PM EST and they close on the Sunday of that week at 11:59 PM (EST). You will have **two opportunities** to submit each quiz. Furthermore, I will open up an **ungraded discussion forum** pertaining to that quiz in which you will be able to discuss with your classmates and the instructor the quiz questions. The purpose of this is to give incentives for a collaborative learning environment. You are not required to post anything in this type of discussion forum but I strongly recommend that you participate, since learning is not only done through ourselves but also through our classmates and peers.

7.1.3 Final Exam (35%)

The final exam is a one-sit timed test. Once you start it, you will have 3 hours to complete it (one-sit). It will open on July 6th at 12:00 PM EST and will close on July 7th at 8:00 PM EST. Topics will include all readings, quizzes and discussion forums covered throughout the course. You will have **only one opportunity** to submit the exam. Since this is an online exam, you are allowed to consult any material you want. The exam will be worth 35 points toward your final grade.

7.1.4 Late/Missed Assignments Policy

Important: If at any point during the exam or quizzes you should encounter any technical issues, please send an email to the help desk at spd_online@stonybrook.edu; You should also copy me on this email so that I am aware of the situation and if possible include a screen-shot of the issue. No make up's are available beyond technical issues that do not include time-out of the exam. You should pace yourself accordingly.

7.1.5 Grades and Grading Scale

Assignment of letter grades is based on the total amount you obtained in all course assignments. Your numerical grade will be converted to a letter grade based on the following table.

Letter Grade	Score Achieved
A	[95, 100]
A–	[90, 95)
B+	[87, 90)
B	[83, 87)
B–	[77, 83)
C+	[70, 77)
C	[63, 70)
C–	[57, 63)
D+	[50, 57)
D	[40, 50)
F	[0, 40)

8 Academic Policies

Academic Integrity Statement

Each student must pursue his or her academic goals honestly and be personally accountable for all submitted work. Representing another person's work as your own is always wrong. Faculty are required to report any suspected instances of academic dishonesty to the Academic Judiciary. Faculty in the Health Sciences Center

(School of Health Technology & Management, Nursing, Social Welfare, and Dental Medicine) and School of Medicine are required to follow their school-specific procedures. For more comprehensive information on academic integrity, including categories of academic dishonesty, please refer to the academic judiciary website at <http://www.stonybrook.edu/uaa/academicjudiciary/>

8.1 Critical Incident Management

Stony Brook University expects students to respect the rights, privileges, and property of other people. Faculty are required to report to the Office of Judicial Affairs any disruptive behavior that interrupts their ability to teach, compromises the safety of the learning environment, or inhibits students' ability to learn. Faculty in the HSC Schools and the School of Medicine are required to follow their school-specific procedures.

8.2 University Student Conduct Code

It can be found at: <http://studentaffairs.stonybrook.edu/ucs/docs/universitystudentconductcode.pdf>

8.3 ADA & Disability Support Services (DSS) Statement

The Rehabilitation Act of 1973 - Section 504 applies to all postsecondary educational programs that receive federal assistance. Reasonable accommodations and academic assistance are provided to students with disabilities registered with the Disability Support Services, ECC (Educational Communications Center) Building, room 128, (631) 632-6748. They will determine with you what accommodations, if any, are necessary and appropriate. All information and documentation is confidential. For procedures and information go to the following website: <http://www.stonybrook.edu/ehs/fire/disabilities>

8.4 Course Materials and Copyright Statement

Course material accessed from Blackboard, SB Connect, SB Capture or a Stony Brook Course website is for the exclusive use of students who are currently enrolled in the course. Content from these systems cannot be reused or distributed without written permission of the instructor and/or the copyright holder. Duplication of materials protected by copyright, without permission of the copyright holder is a violation of the Federal copyright law, as well as a violation of Stony Brook's Academic Integrity and Student Conduct Codes.

9 Communication Competency

You convey a certain image of yourself every time you express yourself-whether it's through the written or spoken word. As such, successful professionals (students, managers, employees) must have effective written and oral communication skills. Therefore, communication errors (such as improper grammar, improper sentence and paragraph structure, misspelling and incorrect punctuation) are unacceptable in coursework throughout. Faculty members also consider communication competency when evaluating student performance in addition to coursework.

9.1 Email and Internet

You must have an active Stony Brook University e-mail account and access to the Internet. All instructor correspondence will be sent to your SBU e-mail account. Please plan on checking your SBU email account regularly for course related messages. To log in to Stony Brook Google Mail, go to <http://www.stonybrook.edu/mycloud> and sign in with your NetID and password.

10 Net-Etiquette Guidelines

The following guidelines for participation in the Threaded Discussion Area and Chat rooms.

- Remember that the absence of face-to-face communication it's easy to misunderstand what is being said;
- Carefully review and read materials that you receive electronically to ensure that you fully understand the message;
- Be sure to carefully re-read and understand what you will be sending in order to ensure that you are not misunderstood by anyone;
- Avoid cluttering your messages with excessive emphasis (stars, arrows, exclamations);
- If you are responding to a message, either include the relevant part of the original message in your message, or make sure refer to the original's contents so as to avoid confusion;
- Be specific and clear, especially when asking questions;
- If your messages can be typed in UPPER and lower case, please use the two appropriately instead of all UPPERCASE characters. This gives the appearance of shouting and makes the message less readable;
- Remember that not all readers have English as their native language, so make allowance for possible misunderstandings and unintended discourtesies;
- Do not abuse new users of computer networks for their lack of knowledge;
- Follow the same standards of politeness as you do in any other aspect of your life.

11 Technical Requirements

This course requires that you have access to the Internet. You are responsible for having a reliable computer and internet connection throughout the course. You will need to have access to, and be able to use the following software:

- A web browser (for instance, Google Chrome, Mozilla Firefox or Internet Explorer)
- Adobe Acrobat Reader (free)
- Adobe Flash Player (free)
- Microsoft Word (free to Stony Brook University students, go to <https://it.stonybrook.edu/software/title/microsoft-office>)

Please note! You **will** be limited if you expect to complete your work on a smart phone or tablet. It will not be possible to submit the Microsoft Word files required for your quiz/exam assignments, neither to view some of the embedded material in Blackboard. This course uses Blackboard for the facilitation of communications between faculty and students, submission of assignments, and posting of grades. The Blackboard Course Site can be accessed at <https://blackboard.stonybrook.edu>.

12 Getting Technical Help

12.1 Campus Network or Blackboard Outage

When access to Blackboard is not available for an extended period of time (greater than one entire evening—6PM till 11PM EST) you can reasonably expect that the due date for assignments will be changed to the next day (assignment still due by midnight).

12.2 Help-Desk

Go to the menu item “SPD Online Support” and go to the Discussion Board within the support site. There you will find a Technical Questions and Answers Forum. Post your question there, and someone from the SPD Online office will respond. Alternatively, you could call the SPD Online office at 631.632.9484 between the hours of 9:00am and 5:00pm, Monday through Friday or e-mail spd_online@stonybrook.edu. For assistance after 5:00 PM EST or over the weekend, please contact the Open SUNY Technical Support, information can be found at <http://open.suny.edu/support/contact-us/current-students/>

12.3 Getting Help with Bb Learning Management System (LMS)

First contact SPD Online at spd_online@stonybrook.edu or 631-632-9484. After hours contact Open SUNY Technical Support, see above for contact information. Additionally students that need help with Bb can contact the TLT Student Help Desk by calling (631) 632-9602, emailing helpme@stonybrook.edu; more information is available via Stony Brook IT: <http://it.stonybrook.edu/services/blackboard#section-6706>. Frequently ask questions about the Bb LMS along with tutorials are available here: <http://it.stonybrook.edu/services/blackboard/navigate-manage>.

13 Academic Support

13.1 SPD Subject Guides

The SPD Subject Guide is a library website specifically designed to assist SPD students with their research. It contains all of the information referenced on this Blackboard page as well as recommendations for specific databases and a live librarian chat feature. Take a look: <http://guides.library.stonybrook.edu/spd>

13.2 Student Support for Online Learning

<http://www.stonybrook.edu/commcms/onlineed/student.html>

13.3 Library Instruction Website

<http://library.stonybrook.edu/workshops-this-week-citation-skills-worldcat-and-endnote-the-hsc/>

13.4 SBU Library Research Guides and Tutorials

<http://library.stonybrook.edu/research/research-basics/>

14 Course Schedule

This weekly schedule is designed in a way that supports students with the time allocation and also gives them a benchmark of progress. Feel free to work at your own pace, but make sure you meet all deadlines timely. As explained in the Course Description section above, this course is divided into **Six Modules**, one for each week, as follows:

- **Module 1** (May 28–June 2)
 - Decision under uncertainty.
 - Introductory forum and Module topic forum.
 - Quiz 1: Online
 - * Open: Friday May 31, 12:00PM EST
 - * Close: Sunday June 2, 11:59PM EST

- **Module 2** (June 3–June 9)
 - Dominant strategies & Strategic Dominance relationship. Strategic Equilibrium, first half.
 - Module topic forum.
 - Quiz 2: Online
 - * Open: Friday June 7, 12:00PM EST
 - * Close: Sunday June 9, 11:59PM EST
- **Module 3** (June 10–June 16)
 - Strategic Equilibrium, second half.
 - Module topic forum.
 - Quiz 3: Online
 - * Open: Friday June 14, 12:00PM EST
 - * Close: Sunday June 16, 11:59PM EST
- **Module 4** (June 17–June 23)
 - Dynamic Competition & Auctions
 - Module topic forum.
 - Quiz 4: Online
 - * Open: Friday June 21, 12:00PM EST
 - * Close: Sunday June 23, 11:59PM EST
- **Module 5** (June 24–June 30)
 - Auctions & Stable matches
 - Module topic forum.
 - Quiz 5: Online
 - * Open: Friday June 28, 12:00PM EST
 - * Close: Sunday June 30, 11:59PM EST
- **Module 6** (July 1–July 6)
 - Stable matches in two-sided markets.
 - Module topic forum.
 - Quiz 6: Online
 - * Open: Wednesday July 3, 12:00PM EST
 - * Close: Thursday July 4, 11:59PM EST
 - **Final Exam:** Online
 - * Open: Friday July 5, 12:00PM, EST
 - * Close: Saturday July 6 (Saturday), 8:00pm, EST
 - * One-sit exam: once you start your exam, you will have 3h to complete it

15 Disclaimer

Subsequent changes can be made to any part of this syllabus when necessary. Any changes will be announced in class and on BlackBoard.